Notes: Kashyap and Stein (AER, 2000)

What Do a Million Observations on Banks Say About the Transmission of Monetary Policy?

Contents

1 Big picture

- **Question.** Does monetary policy affect real bank loan supply, or do observed declines in lending after monetary contractions simply reflect weaker loan demand?
- **Core idea.** The paper tests the **bank lending channel**: if monetary tightening drains reservable insured deposits, and if banks cannot perfectly replace those deposits with uninsured nonreservable liabilities, then some banks must contract assets. Lending should fall most at banks with weak liquidity buffers.
- **Key empirical design.** Instead of looking only at aggregate lending, the paper uses almost one million bank-quarter observations for every insured U.S. commercial bank from 1976Q1 to 1993Q2. It asks whether lending becomes more dependent on bank liquidity during tight-money episodes.
- **Main result.** Lending at less-liquid banks responds more strongly to monetary policy, especially among small banks. This is exactly the cross-sectional pattern predicted by a bank lending channel.
- **Why small banks matter.** Small banks are less likely to have frictionless access to uninsured finance such as large CDs, federal funds borrowing, or other market liabilities. They should therefore be more constrained after policy tightening.
- **Loan category distinction.** Results are strongest for commercial and industrial loans. Results for total loans have the same general sign pattern but are weaker, likely because total loans combine loan categories with different cyclicality and maturity.
- **Economic takeaway.** The paper does not precisely estimate the aggregate macro importance of the lending channel, but it makes the existence of a lending channel difficult to deny for the United States during the sample period.

2 Incremental contribution of the paper

- **Uses bank-level heterogeneity to identify loan supply.** Earlier aggregate evidence could be explained either by loan-supply contractions or by borrowers demanding fewer loans. This paper uses cross-bank differences in liquidity to test whether the same monetary shock has different effects on banks that should differ in their ability to protect lending.
- **Connects the lending channel to bank balance-sheet liquidity.** The central test is not merely whether small-bank lending falls after tightening. It is whether, within size classes, lending is more sensitive to monetary policy for banks with lower securities-to-assets ratios.
- **Separates size from liquidity.** The authors argue that large banks are more likely to raise uninsured funds, so the liquidity interaction should be strongest among small banks and weaker or absent among the largest banks.
- **Uses an unusually large regulatory panel.** The data set contains quarterly Call Report observations for all insured U.S. commercial banks over nearly two decades, allowing precise cross-sectional comparisons that were not possible in aggregate time-series work.
- **Explicitly confronts endogeneity of bank liquidity.** Liquidity is not randomly assigned. The paper discusses the direction of possible biases and uses patterns across bank-size classes,

GDP controls, one-step specifications, subsamples, and quasi-IV checks to argue that the main finding is not a spurious loan-demand artifact.

- **Provides a quantitative benchmark.** The authors translate their coefficients into differences in C&I lending between liquid and illiquid small banks after a 100-basis-point funds-rate shock, and then compare the implied effect with aggregate small-bank lending movements.

3 The bank lending channel and hypotheses

3.1 The mechanism

The bank lending view rests on a failure of the Modigliani-Miller proposition for banks. In a frictionless world, a bank that loses reservable insured deposits after monetary tightening could replace them one-for-one with nonreservable uninsured liabilities. In that case, open-market operations would change the composition of bank liabilities but not bank loan supply.

The lending-channel logic is different:

- monetary tightening drains reserves and reduces insured deposit funding;
- uninsured funding sources are costly or rationed because they are not protected by deposit insurance and may be subject to adverse-selection problems;
- banks with limited access to replacement funds must shrink assets;
- banks with large securities portfolios can protect lending by drawing down liquid assets;
- banks with small liquidity buffers must cut loans more aggressively.

3.2 Main hypotheses

Let L_{it} denote lending by bank i, B_{it} denote balance-sheet liquidity, and M_t denote the stance of monetary policy, with higher M_t representing easier policy. The first hypothesis is:

$$\frac{\partial^2 L_{it}}{\partial B_{it} \partial M_t} < 0. \tag{1}$$

This means that liquidity matters more when policy is tight. Equivalently, as policy becomes easier, the cross-sectional dependence of loan growth on bank liquidity should weaken.

The second hypothesis adds bank size:

$$\frac{\partial^3 L_{it}}{\partial B_{it} \partial M_t \partial SIZE_{it}} > 0. \tag{2}$$

Because large banks have better access to uninsured finance, the liquidity-policy interaction should be strongest for small banks and weaker for very large banks.

3.3 Interpretation in words

- **Within small banks,** an illiquid bank should reduce lending more after a monetary contraction than a liquid bank.
- **Across size classes,** this liquidity effect should be concentrated among small banks. Large banks should be less dependent on internal liquidity because they can issue or borrow in uninsured markets more easily.

- **The key test is interactive.** The paper is not just asking whether monetary tightening reduces bank lending. It asks whether the effect of tightening is systematically different across banks with different liquidity positions.

4 Data and measurement

4.1 Bank-level data

- **Source.** Consolidated Reports of Condition and Income, commonly called Call Reports.
- **Coverage.** Every insured U.S. commercial bank reporting quarterly balance-sheet and income-statement data from 1976Q1 to 1993Q2.
- **Raw size.** 961,530 bank-quarter observations.
- **Cleaning.** The authors remove merger-quarter observations, extreme loan-growth outliers, and observations without enough consecutive loan-growth history.
- **Regression samples.** The main C&I loan regressions use 746,179 observations. The total-loan regressions use 836,885 observations.

4.2 Bank-size classes

The sample is split into three size groups:

Size class	Interpretation
Below 95th percentile	Small banks
95th to 99th percentile	Medium/large banks
Above 99th percentile	Very large banks

This split reflects the highly skewed distribution of bank assets. Even many banks below the 95th percentile are small by national capital-market standards.

4.3 Loan variables

- **Total loans.** Broad lending measure including real estate, C&I, consumer, and other loans.
- **C&I loans.** Commercial and industrial lending. These loans are especially relevant for the bank lending channel and are likely to adjust faster than many long-term loan categories.
- **C&I sample screen.** In C&I regressions, the authors omit banks for which C&I loans are less than 5 percent of total loans.

4.4 Liquidity variable

The liquidity measure B_{it} is:

$$B_{it} = \frac{\text{securities}_{it} + \text{federal funds sold}_{it}}{\text{assets}_{it}}. \qquad (3)$$

- Securities and federal funds sold are assets that can be drawn down to protect lending.

- Cash is excluded because much of it may reflect required reserves that cannot be freely used as a buffer.
- Robustness checks show that including cash does not materially change results.

4.5 Monetary-policy indicators

The paper uses three measures of monetary policy. All are oriented so that higher values correspond to easier policy.

1. **Boschen-Mills index.** A narrative measure based on FOMC documents. The authors code the stance of policy from strongly expansionary to strongly contractionary.
2. **Federal funds rate.** The sign is inverted so that higher values mean easier policy. The authors note that the funds rate may be less reliable during the Volcker operating-procedure period.
3. **Bernanke-Mihov indicator.** A VAR-based indicator designed to flexibly account for changes in Federal Reserve operating procedures.

5 Empirical design

5.1 Two-step baseline strategy

The baseline empirical design is intentionally flexible. The authors estimate a separate first-stage cross-sectional regression for each quarter and size class:

$$\Delta \log(L_{it}) = \sum_{j=1}^{4} \alpha_{tj} \Delta \log(L_{i,t-j}) + \beta_t B_{i,t-1} + \sum_{k=1}^{12} \gamma_{kt} FRB_{ik} + \epsilon_{it}. \tag{4}$$

- L_{it} is either C&I loans or total loans.
- Lagged loan growth controls for bank-level persistence in lending.
- Federal Reserve district dummies control for geography.
- The key output is β_t, the quarter-specific relationship between bank liquidity and loan growth within a size class.

The second step asks whether β_t varies systematically with monetary policy. The univariate specification is:

$$\beta_t = \eta + \sum_{j=0}^{4} \phi_j \Delta M_{t-j} + \delta TIME_t + u_t. \tag{5}$$

The bivariate specification adds current and lagged GDP growth:

$$\beta_t = \eta + \sum_{j=0}^{4} \phi_j \Delta M_{t-j} + \sum_{j=0}^{4} \gamma_j \Delta GDP_{t-j} + \delta TIME_t + u_t. \tag{6}$$

The paper tests whether $\sum_j \phi_j < 0$ for small banks. If easier policy reduces the importance of internal liquidity, the estimated coefficient of liquidity on loan growth should fall when policy loosens.

5.2 Why use a two-step method?

- **Demand controls are very flexible.** The first step effectively absorbs any quarterly macro shock common to all banks in a Federal Reserve district within a size class.
- **Identification comes from within-period cross-bank variation.** The method asks whether more-liquid and less-liquid banks behave differently in the same quarter and size group.
- **Trade-off.** The method sacrifices statistical power relative to a pooled one-step interaction regression, but it is more conservative with respect to unobserved loan-demand shocks.

5.3 Endogeneity of bank liquidity

Bank liquidity is endogenous. The paper discusses two main bias stories.

1. **Heterogeneous risk aversion.** Conservative banks may both hold more securities and lend to less cyclical borrowers. This would make liquid banks less cyclical for reasons unrelated to funding constraints and could bias the policy-liquidity interaction too negative.
2. **Rational buffer-stocking.** Banks with more cyclically sensitive loan opportunities may rationally hold more liquidity. This would bias estimates toward zero or toward the wrong sign, making the authors' tests conservative.

The paper argues that the data support the second bias story more than the first. In the bivariate regressions, GDP-growth coefficients tend to be positive, and the largest banks often have positive policy coefficients. Both patterns are more consistent with rational buffer-stocking than with heterogeneous risk aversion.

5.4 Bank-capital shock alternative

A different story is that monetary tightening hurts bank capital by creating loan losses, and weakly capitalized banks then cut lending. The paper argues that this story would predict two patterns:

- adding GDP growth should substantially weaken the monetary-policy coefficients;
- GDP coefficients should be negative.

The evidence does not support these predictions. Therefore, the main results are not naturally explained by a bank-capital shock mechanism.

6 Main empirical results

6.1 C&I lending results

The strongest results are for C&I loans.

- For small banks, all six estimates of $\sum_j \phi_j$ are negative across the three monetary measures and two second-step specifications.
- The federal-funds-rate estimates are especially precise: for small banks, the univariate C&I estimate is -0.0267 and the bivariate estimate is -0.0151.
- Big-bank estimates are positive in every C&I specification. This creates large negative small-bank minus big-bank differentials.

- Each of the six C&I small-big differentials is statistically significant at conventional levels; four have very small p-values.

Interpretation. Among small banks, lending depends more on liquid-asset buffers when money is tight. Very large banks do not show the same pattern, consistent with their better access to uninsured funding markets.

6.2 Total-loan results

The total-loan results point in the same direction but are weaker.

- Five of six small-bank estimates are negative.
- Big-bank estimates are positive in all six cases.
- The magnitude of the small-bank estimates is often only one-third to one-quarter of the corresponding C&I estimates.

Interpretation. Total loans combine different loan categories with different maturities and different cyclical sensitivities. C&I loans are likely to be the cleaner and faster-adjusting category for detecting a lending-channel effect.

6.3 Detailed lag patterns

The full second-step details show that the findings are not driven by a single odd lag.

- For small banks and C&I loans using the federal-funds-rate measure, nearly all individual lag coefficients on monetary policy are negative.
- The response has a plausible hump-shaped pattern: effects build over the first few lags and then decline.
- GDP-growth coefficients are mostly positive in the bivariate versions, which supports the rational buffer-stocking interpretation of the remaining bias.

7 Robustness checks

7.1 One-step interaction specification

The authors also estimate pooled one-step models where loan growth is regressed directly on liquidity, monetary policy, and interactions between liquidity and monetary-policy changes. These models are more tightly parametrized than the two-step approach.

- Point estimates are generally close to the two-step estimates.
- Standard errors are much smaller.
- Statistical significance becomes stronger.

Read: the one-step results show that the conservative two-step approach is not producing an artificial finding; if anything, using more structure makes the evidence stronger.

7.2 Subsample results

The paper splits the sample into 1976Q1-1985Q4 and 1986Q1-1993Q2.

- Results are much stronger in the later subsample beginning in 1986Q1.
- In the later period, most C&I test statistics have the expected sign and are statistically significant despite the smaller sample.
- This reduces concern that the results are driven by the Volcker operating-procedure period or by Regulation Q restrictions.

7.3 Additional robustness mentioned in the paper

The main conclusions are not materially changed by:

- aggregating banks to the bank holding company level;
- adding cash to the liquidity measure;
- using alternative outlier screens;
- using tighter geographic controls;
- using more complex lag structures;
- excluding the time trend;
- using quasi-instruments that purge liquidity of correlations with observable loan-risk composition.

8 Economic magnitude

8.1 Liquid versus illiquid small banks

Using the conservative bivariate federal-funds-rate estimate for small-bank C&I loans, the authors compare two equal-sized small banks:

- a liquid bank at the 90th percentile of small-bank liquidity, with $B_{it} = 60.2\%$;
- an illiquid bank at the 10th percentile, with $B_{it} = 20.6\%$.

After a 100-basis-point increase in the funds rate, the illiquid bank's C&I loans are predicted to be roughly 0.6 percent lower than the liquid bank's C&I loans after four quarters. Using the more aggressive small-bank minus big-bank differential, the predicted gap is about 5.3 percent.

8.2 Aggregate small-bank lending

Integrating over the distribution of small-bank liquidity, the conservative estimate implies that, after a 100-basis-point funds-rate shock, total C&I lending by small banks is 0.41 percent lower than it would be if all small banks were unconstrained. The more aggressive estimate based on small-bank minus big-bank differentials implies a 3.62 percent reduction.

8.3 How large is this?

The conservative estimate explains about 12 percent of the small-bank aggregate C&I lending decline found in earlier Kashyap-Stein work. The more aggressive estimate can explain the full aggregate small-bank C&I decline. Because the aggressive estimate relies on treating big-bank coefficients as pure bias corrections, the authors are cautious about using it as a precise macro estimate.

9 Figures and tables

9.1 Table 1: Bank balance sheets by size

Read:

- Small banks hold much larger shares of assets in securities and have simpler liability structures.
- The largest banks hold fewer securities and use more nondeposit liabilities such as federal funds borrowing and other liabilities.
- These cross-sectional patterns fit the idea that small banks need larger buffer stocks because they have less access to uninsured external finance.

Economic message: Table 1 motivates the size split. Small banks look balance-sheet constrained in a way that large banks do not.

TABLE 1—BALANCE SHEETS FOR BANKS OF DIFFERENT SIZES

Panel A: Composition of Bank Balance Sheets as of 1976 Q1

	Below 75th percentile	Between 75th and 90th percentile	Between 90th and 95th percentile	Between 95th and 98th percentile	Between 98th and 99th percentile	Above 99th percentile
Number of banks	10,784	2,157	719	431	144	144
Mean assets (1993 $ millions)	32.82	119.14	247.73	556.61	1,341.45	10,763.44
Median assets (1993 $ millions)	28.43	112.63	239.00	508.06	1,228.66	3,964.55
Fraction of total system assets	0.13	0.09	0.06	0.09	0.07	0.56
Fraction of total assets in size category						
Cash	0.09	0.09	0.10	0.12	0.13	0.22
Securities	0.34	0.33	0.32	0.29	0.27	0.15
Federal funds lent	0.05	0.04	0.04	0.05	0.04	0.03
Total domestic loans	0.52	0.53	0.53	0.53	0.54	0.41
Real estate loans	0.17	0.19	0.20	0.18	0.17	0.09
C & I loans	0.10	0.13	0.15	0.16	0.17	0.17
Loans to individuals	0.15	0.16	0.15	0.15	0.14	0.06
Total deposits	0.90	0.90	0.89	0.87	0.84	0.81
Demand deposits	0.31	0.30	0.30	0.31	0.33	0.25
Time and savings deposits	0.59	0.60	0.59	0.55	0.51	0.33
Time deposits > $100K	0.07	0.10	0.12	0.14	0.14	0.16
Federal funds borrowed	0.00	0.01	0.02	0.04	0.07	0.08
Subordinated debt	0.00	0.00	0.00	0.00	0.01	0.01
Other liabilities	0.01	0.01	0.01	0.01	0.02	0.06
Equity	0.08	0.08	0.07	0.07	0.07	0.05

Panel B: Composition of Bank Balance Sheets as of 1993 Q2

	Below 75th percentile	Between 75th and 90th percentile	Between 90th and 95th percentile	Between 95th and 98th percentile	Between 98th and 99th percentile	Above 99th percentile
Number of banks	8,404	1,681	560	336	112	113
Mean assets (1993 $ millions)	44.42	165.81	380.14	1,072.57	3,366.01	17,413.41
Median assets (1993 $ millions)	38.59	155.73	362.75	920.78	3,246.33	9,297.70
Fraction of total system assets	0.10	0.08	0.06	0.10	0.11	0.55
Fraction of total assets in size category						
Cash	0.05	0.05	0.05	0.07	0.07	0.09
Securities	0.34	0.32	0.29	0.27	0.25	0.22
Federal funds lent	0.04	0.04	0.03	0.04	0.04	0.04
Total loans	0.53	0.56	0.60	0.59	0.60	0.59
Real estate loans	0.30	0.33	0.34	0.30	0.25	0.21
C & I loans	0.09	0.10	0.11	0.12	0.13	0.18
Loans to individuals	0.09	0.10	0.12	0.14	0.17	0.10
Total deposits	0.88	0.87	0.85	0.79	0.76	0.69
Transaction deposits	0.26	0.26	0.25	0.24	0.26	0.19
Large deposits	0.17	0.21	0.22	0.25	0.24	0.21
Brokered deposits	0.00	0.00	0.01	0.02	0.02	0.01
Federal funds borrowed	0.01	0.02	0.04	0.06	0.10	0.09
Subordinated debt	0.00	0.00	0.00	0.00	0.00	0.02
Other liabilities	0.01	0.02	0.03	0.05	0.06	0.13
Equity	0.10	0.09	0.08	0.09	0.08	0.07

Table 1: Balance sheets for banks of different sizes.

Small banks hold more liquid securities and rely more heavily on deposits and equity; large banks have broader funding options.

9.2 Figure 1 and Table 2: Monetary-policy indicators

Figure 1 read:

- The Boschen-Mills index, Bernanke-Mihov indicator, and inverted federal funds rate broadly agree on major policy episodes.
- All three identify very tight policy around the 1979 operating-procedure change, loose policy around 1985-1986, and renewed tightening around 1988.

Table 2 read:

- The indicators are moderately correlated in levels over the full sample.
- Correlations are weaker in the first half of the sample, especially at quarterly frequencies.
- Correlations are much higher in the second half, suggesting cleaner policy measurement after 1986.

Economic message: The paper does not rely on a single monetary-policy measure. The main results survive across very different policy indicators.

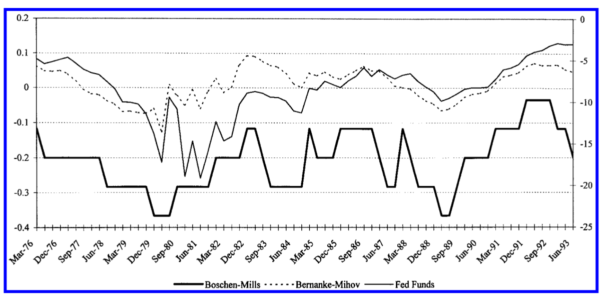

FIGURE 1. MEASURES OF MONETARY POLICY

(a) Figure 1: Measures of monetary policy.

The three measures capture similar broad monetary-policy episodes.

TABLE 2—CORRELATIONS OF MEASURES OF MONETARY POLICY

	Correlation of:		
	Levels	Annual changes	Quarterly changes
A. Full sample (76Q1–93Q2)			
1. Boschen-Mills/Federal funds	0.608	0.382	0.219
2. Boschen-Mills/ Bernanke-Mihov	0.710	0.416	0.099
3. Federal funds/ Bernanke-Mihov	0.580	0.486	0.483
B. 1st-half sample (76Q1–85Q4)			
1. Boschen-Mills/Federal funds	0.514	0.318	0.233
2. Boschen-Mills/ Bernanke-Mihov	0.665	0.293	0.018
3. Federal funds/ Bernanke-Mihov	0.526	0.476	0.471
C. 2nd-half sample (86Q1–93Q2)			
1. Boschen-Mills/Federal funds	0.733	0.647	0.414
2. Boschen-Mills/ Bernanke-Mihov	0.844	0.734	0.361
3. Federal funds/ Bernanke-Mihov	0.871	0.567	0.730

Notes: Annual changes are defined as the change between the level of a variable in a certain quarter and the level four quarters before that. The sign of the federal funds rate has been inverted to preserve the convention in the paper that a higher level of the monetary-policy measure reflects a looser policy.

(b) Table 2: Correlations of monetary-policy measures.

Correlations are strongest in the later sample period and weaker during the volatile Volcker years.

9.3 Table 3: Main two-step results

Read:

- For C&I loans, small-bank coefficients are negative across all three monetary indicators and both univariate and bivariate specifications.
- Big-bank coefficients are positive, so small-bank minus big-bank differences are large and negative.
- For total loans, the signs are similar but estimates are smaller and less precise.

Economic message: Table 3 is the paper's central result. Monetary policy affects lending more strongly at illiquid small banks, exactly as a bank lending channel predicts.

Table 3—Two-Step Estimation of Equations (1), (2), and (3): Sum of Coefficients on Monetary-Policy Indicator

Panel A: C&I Loans	Univariate	Bivariate
1. Boschen-Mills		
<95	−0.0438	−0.0131
	(0.0188)	(0.0187)
95–99	−0.0339	0.0094
	(0.0401)	(0.0303)
>99	0.0960	0.1411
	(0.0661)	(0.0428)
Small-Big	−0.1398	−0.1542
	(0.0611)	(0.0449)
2. Funds rate		
<95	−0.0267	−0.0151
	(0.0071)	(0.0089)
95–99	−0.0066	0.0097
	(0.0137)	(0.0112)
>99	0.0795	0.1175
	(0.0281)	(0.0314)
Small–Big	−0.1062	−0.1327
	(0.0296)	(0.0376)
3. Bernanke-Mihov		
<95	−1.8633	−0.5269
	(1.0933)	(1.2463)
95–99	0.7345	3.3461
	(2.1853)	(2.1119)
>99	4.7862	7.5911
	(3.5220)	(2.3927)
Small–Big	−6.6495	−8.1181
	(3.3966)	(3.0215)

Panel B: Total Loans	Univariate	Bivariate
1. Boschen-Mills		
<95	−0.0179	−0.0044
	(0.0110)	(0.0120)
95–99	−0.0129	0.0167
	(0.0236)	(0.0118)
>99	0.0516	0.0921
	(0.0522)	(0.0373)
Small–Big	−0.0695	−0.0965
	(0.0464)	(0.0348)
2. Funds rate		
<95	−0.0088	−0.0046
	(0.0037)	(0.0049)
95–99	−0.0126	−0.0040
	(0.0079)	(0.0060)
>99	0.0258	0.0460
	(0.0188)	(0.0152)
Small–Big	−0.0346	−0.0506
	(0.0182)	(0.0174)
3. Bernanke-Mihov		
<95	−0.1926	0.7827
	(0.5344)	(0.5780)
95–99	−0.2849	1.1191
	(1.1178)	(0.7766)
>99	3.6558	6.7373
	(2.5209)	(1.4636)
Small–Big	−3.8484	−5.9545
	(2.2180)	(1.5971)

Note: Standard errors are in parentheses.

Table 3: Two-step estimation - sum of coefficients on monetary-policy indicators.

Small-bank C&I loan results are negative; big-bank results are positive, producing strong small-big differentials.

9.4 Table 4: Full lag details

Read:

- Table 4 shows the individual lag coefficients behind the sums reported in Table 3.
- In the federal-funds-rate C&I specification, the small-bank monetary-policy coefficients are consistently negative.
- GDP-growth coefficients are generally positive, which is consistent with rational buffer-stocking rather than the heterogeneous-risk-aversion bias story.

Economic message: The main estimates are not hiding unstable lag patterns. The detailed coefficients generally move in the direction the theory predicts.

TABLE 4—TWO-STEP ESTIMATION OF EQUATIONS (1), (2), AND (3): FULL DETAILS

Panel A: Money Measure: Change in Boschen-Mills

	Monetary-policy indicator					Change in GDP				
C&I loans	0	1	2	3	4	0	1	2	3	4
Univariate										
<95	−0.0074	−0.0066	−0.0138	−0.0137	−0.0023					
(R^2 = 0.1357)	(0.0074)	(0.0041)	(0.0044)	(0.0056)	(0.0048)					
95–99	0.0088	−0.0172	−0.0226	−0.0139	0.0111					
(R^2 = 0.1259)	(0.0139)	(0.0132)	(0.0211)	(0.0132)	(0.0140)					
>99	−0.0193	0.0105	0.0329	−0.0002	0.0722					
(R^2 = 0.0988)	(0.0114)	(0.0189)	(0.0315)	(0.0220)	(0.0205)					
Bivariate										
<95	−0.0037	−0.0008	−0.0064	−0.0054	0.0031	0.6259	0.2955	0.7707	0.9165	0.2920
(R^2 = 0.3404)	(0.0057)	(0.0046)	(0.0033)	(0.0076)	(0.0047)	(0.5292)	(0.2489)	(0.3127)	(0.2665)	(0.3831)
95–99	0.0151	−0.0088	−0.0149	−0.0033	0.0213	0.9987	0.7424	0.6632	0.5668	1.2774
(R^2 = 0.2086)	(0.0133)	(0.0148)	(0.0177)	(0.0107)	(0.0133)	(0.5464)	(0.8997)	(1.2009)	(1.0185)	(0.6125)
>99	−0.0141	0.0100	0.0317	0.0225	0.0910	1.2170	0.8454	−3.8951	4.2036	3.1755
(R^2 = 0.2589)	(0.0088)	(0.0175)	(0.0201)	(0.0146)	(0.0185)	(1.4284)	(1.1834)	(2.8855)	(2.4012)	(1.0001)

Panel B: Money Measure: Change in Federal Funds Rate

	Monetary-policy indicator					Change in GDP				
C&I loans	0	1	2	3	4	0	1	2	3	4
Univariate										
<95	−0.0069	−0.0077	−0.0062	−0.0054	−0.0005					
(R^2 = 0.2868)	(0.0019)	(0.0022)	(0.0020)	(0.0012)	(0.0012)					
95–99	−0.0018	−0.0030	−0.0027	−0.0056	0.0065					
(R^2 = 0.0834)	(0.0021)	(0.0038)	(0.0058)	(0.0054)	(0.0013)					
>99	0.0118	0.0140	0.0235	0.0142	0.0161					
(R^2 = 0.0958)	(0.0075)	(0.0063)	(0.0082)	(0.0042)	(0.0075)					
Bivariate										
<95	−0.0053	−0.0059	−0.0040	−0.0023	0.0024	0.5664	−0.0587	0.4089	1.0498	0.5039
(R^2 = 0.4526)	(0.0014)	(0.0021)	(0.0024)	(0.0026)	(0.0021)	(0.5115)	(0.3923)	(0.3866)	(0.4193)	(0.2334)
95–99	0.0015	0.0006	0.0002	−0.0036	0.0110	1.0847	0.7481	−0.0294	1.3017	1.2965
(R^2 = 0.1870)	(0.0034)	(0.0045)	(0.0052)	(0.0063)	(0.0028)	(0.7850)	(1.1979)	(0.9570)	(0.8977)	(0.6423)
>99	0.0146	0.0113	0.0236	0.0250	0.0431	1.2667	−0.7602	−5.1621	7.5245	3.3015
(R^2 = 0.3442)	(0.0068)	(0.0051)	(0.0103)	(0.0093)	(0.0133)	(1.8068)	(1.6353)	(3.4558)	(2.1000)	(1.6629)

Panel C: Money Measure: Change in Bernanke-Mihov

	Monetary-policy indicator					Change in GDP				
C&I loans	0	1	2	3	4	0	1	2	3	4
Univariate										
<95	−0.0237	0.2379	−0.2093	−1.2548	−0.6135					
(R^2 = 0.1440)	(0.3317)	(0.4173)	(0.3736)	(0.3094)	(0.2980)					
95–99	1.1588	0.9354	−0.3920	−1.5933	0.6255					
(R^2 = 0.0887)	(0.5546)	(0.6414)	(0.9313)	(1.0750)	(0.5074)					
>99	0.0938	0.5484	0.6991	1.7404	1.7045					
(R^2 = 0.0309)	(1.7450)	(1.3596)	(1.5339)	(1.0032)	(1.0018)					
Bivariate										
<95	0.1069	0.1053	−0.1351	−0.6614	0.0574	0.6555	0.1692	0.8708	0.8861	0.3600
(R^2 = 0.3535)	(0.2957)	(0.3275)	(0.2422)	(0.3316)	(0.4119)	(0.6661)	(0.2782)	(0.3824)	(0.2394)	(0.2609)
95–99	1.4562	0.8915	−0.1876	−0.6774	1.8634	1.5949	0.5113	1.3196	0.6149	1.089
(R^2 = 0.2043)	(0.4627)	(0.5309)	(0.8566)	(0.8073)	(0.4651)	(0.6306)	(0.8733)	(1.3575)	(0.8721)	(0.5117)
>99	1.1890	0.7249	−0.3289	2.6111	3.3949	2.5876	0.8679	−3.8538	4.2408	1.8398
(R^2 = 0.1781)	(1.2061)	(0.9098)	(1.8175)	(1.0684)	(0.9044)	(1.8625)	(1.2244)	(3.9985)	(2.7145)	(1.0878)

(a) Table 4, Panels A-C: Full details for C&I loans.

The detailed C&I specifications show the lag structure for each monetary indicator.

TABLE 4—*Continued.*

Panel D: Money Measure: Change in Boschen-Mills

	Monetary-policy indicator					Change in GDP				
Total loans	0	1	2	3	4	0	1	2	3	4
Univariate										
<95	−0.0045	−0.0124	0.0023	0.0012	−0.0045					
(R^2 = 0.2346)	(0.0040)	(0.0033)	(0.0021)	(0.0035)	(0.0044)					
95–99	0.0028	−0.0066	0.0024	0.0005	−0.0121					
(R^2 = 0.0692)	(0.0064)	(0.0046)	(0.0090)	(0.0073)	(0.0049)					
>99	−0.0238	0.0121	0.0213	0.0202	0.0218					
(R^2 = 0.1208)	(0.0118)	(0.0077)	(0.0210)	(0.0155)	(0.0140)					
Bivariate										
<95	−0.0024	−0.0099	0.0049	0.0051	−0.0020	0.3639	0.2310	0.0841	0.4924	0.1157
(R^2 = 0.3216)	(0.0036)	(0.0040)	(0.0022)	(0.0044)	(0.0042)	(0.2315)	(0.2037)	(0.1818)	(0.3089)	(0.3651)
95–99	0.0068	0.0019	0.0087	0.0070	−0.0077	0.2406	1.4108	0.2038	0.9510	−0.2656
(R^2 = 0.2847)	(0.0055)	(0.0033)	(0.0064)	(0.0047)	(0.0037)	(0.2614)	(0.2796)	(0.3785)	(0.4722)	(0.5489)
>99	−0.0182	0.0153	0.0244	0.0364	0.0341	1.1544	0.6884	−1.7162	2.6482	1.6745
(R^2 = 0.2880)	(0.0115)	(0.0095)	(0.0153)	(0.0132)	(0.0110)	(0.9052)	(0.5291)	(1.6450)	(1.0135)	(0.8612)

Panel E: Money Measure: Change in Federal Funds Rate

Total loans	Monetary-policy indicator					Change in GDP				
0	1	2	3	4	0	1	2	3	4	
Univariate										
<95	−0.0033	−0.0032	−0.0015	−0.0001	−0.0006					
(R^2 = 0.1607)	(0.0011)	(0.0013)	(0.0011)	(0.0009)	(0.0010)					
95–99	−0.0042	−0.0038	−0.0023	−0.0018	−0.0005					
(R^2 = 0.0769)	(0.0018)	(0.0016)	(0.0025)	(0.0016)	(0.0021)					
>99	−0.0041	0.0044	0.0102	0.0068	0.0085					
(R^2 = 0.1084)	(0.0037)	(0.0045)	(0.0056)	(0.0036)	(0.0037)					
Bivariate										
<95	−0.0025	−0.0020	−0.0004	0.0003	0.0000	0.1830	0.3422	0.0758	0.3096	0.1688
(R^2 = 0.2202)	(0.0012)	(0.0017)	(0.0015)	(0.0015)	(0.0018)	(0.3502)	(0.3213)	(0.2251)	(0.4556)	(0.3703)
95–99	−0.0021	0.0002	0.0014	−0.0024	−0.0011	−0.0999	1.7631	0.0265	0.5797	−0.1162
(R^2 = 0.2665)	(0.0011)	(0.0011)	(0.0023)	(0.0018)	(0.0021)	(0.4222)	(0.4076)	(0.3354)	(0.5349)	(0.5732)
>99	−0.0021	0.0034	0.0101	0.0115	0.0231	0.8456	−0.0946	−2.9525	3.7976	2.2407
(R^2 = 0.3450)	(0.0030)	(0.0038)	(0.0054)	(0.0056)	(0.0057)	(1.0891)	(0.6442)	(1.9395)	(0.0745)	(0.9924)

Panel F: Money Measure: Change in Bernanke-Mihov

	Monetary-policy indicator					Change in GDP				
Total loans	0	1	2	3	4	0	1	2	3	4
Univariate										
<95	0.2491	0.0389	0.0671	−0.4285	−0.1192					
(R^2 = 0.1254)	(0.2014)	(0.3583)	(0.1698)	(0.2176)	(0.1805)					
95–99	0.4467	0.4244	0.1109	−1.1053	−0.1616					
(R^2 = 0.1280)	(0.2725)	(0.2800)	(0.3769)	(0.3610)	(0.4164)					
>99	−0.7211	1.0204	2.3224	0.2335	0.8005					
(R^2 = 0.1406)	(1.3088)	(0.6189)	(0.8896)	(0.8930)	(0.7678)					
Bivariate										
<95	0.3874	0.0854	0.1358	−0.1057	0.2799	0.6243	0.3872	0.138	0.2682	0.1502
(R^2 = 0.2420)	(0.1781)	(0.2625)	(0.1331)	(0.1916)	(0.2215)	(0.2974)	(0.1333)	(0.2239)	(0.3109)	(0.3757)
95–99	0.5088	0.5294	0.3072	−0.5354	0.309	0.4486	1.3427	0.0998	0.6261	−0.0366
(R^2 = 0.3040)	(0.2571)	(0.2064)	(0.2425)	(0.3127)	(0.3746)	(0.2477)	(0.2580)	(0.3455)	(0.4909)	(0.5315)
>99	0.0652	1.2694	2.0188	1.079	2.3049	2.585	0.9495	−1.4611	1.6821	1.439
(R^2 = 0.3068)	(1.0846)	(0.4983)	(0.9654)	(1.0308)	(0.3927)	(0.6407)	(0.5772)	(2.0710)	(1.3014)	(0.7247)

Notes: Standard errors are in parentheses. All regressions also contain a time trend, which is not shown.

(b) Table 4, Panels D-F: Full details for total loans.

Total-loan results are directionally similar but weaker than C&I results.

9.5 Table 5 and Table 6: Robustness

Table 5 read:

- The one-step interaction method produces estimates similar to the two-step method.
- The estimates become more precise because the specification imposes more structure and uses more variation.

Table 6 read:

- The later subsample, 1986Q1-1993Q2, produces stronger results.
- This suggests that the main findings are not driven by Regulation Q or by unusual Volcker-era operating procedures.

Economic message: The main result is robust. It becomes stronger in the cleaner later monetary-policy environment.

TABLE 5—ONE-STEP ESTIMATION OF EQUATIONS (4) AND (5): SUM OF COEFFICIENTS ON MONETARY-POLICY INDICATOR

Panel A: C&I Loans	Univariate	Bivariate
1. Boschen-Mills		
<95	−0.0614	−0.0430
	(0.0069)	(0.0077)
95–99	−0.0171	0.0242
	(0.0276)	(0.0288)
>99	0.0862	0.1337
	(0.0530)	(0.0581)
Small–Big	−0.1475	−0.1767
	(0.0535)	(0.0586)
2. Funds rate		
<95	−0.0339	−0.0238
	(0.0022)	(0.0041)
95–99	−0.0013	0.0102
	(0.0133)	(0.0147)
>99	0.0602	0.0903
	(0.0239)	(0.0266)
Small–Big	−0.0941	−0.1141
	(0.0240)	(0.0269)
3. Bernanke-Mihov		
<95	−2.7518	−1.7802
	(0.3920)	(0.4567)
95–99	1.3557	3.7417
	(1.5929)	(1.7226)
>99	3.8509	6.3203
	(2.7288)	(3.0582)
Small–Big	−6.6027	−8.1005
	(2.7568)	(3.0921)

Panel B: Total Loans	Univariate	Bivariate
1. Boschen-Mills		
<95	−0.0309	−0.0268
	(0.0022)	(0.0022)
95–99	−0.0029	0.0392
	(0.0155)	(0.0164)
>99	−0.0117	0.0379
	(0.0336)	(0.0380)
Small–Big	−0.0191	−0.0647
	(0.0337)	(0.0380)
2. Funds rate		
<95	−0.0144	−0.0119
	(0.0007)	(0.0009)
95–99	−0.0013	0.0056
	(0.0077)	(0.0084)
>99	0.0297	0.0509
	(0.0130)	(0.0141)
Small–Big	−0.0440	−0.0628
	(0.0130)	(0.0141)
3. Bernanke-Mihov		
<95	−0.6628	0.0803
	(0.1410)	(0.1626)
95–99	0.4789	2.5306
	(0.8378)	(0.8774)
>99	1.6325	5.3095
	(1.5571)	(1.8224)
Small–Big	−2.2953	−5.2292
	(1.5634)	(1.8296)

(a) Table 5: One-step estimation.

The one-step method confirms the two-step results with smaller standard errors.

TABLE 6—TWO-STEP ESTIMATION OF EQUATIONS (1), (2), AND (3): SPLIT SAMPLE RESULTS: SUM OF COEFFICIENTS ON MONETARY-POLICY INDICATOR

Panel A: C&I Loans	76Q1–85Q4		86Q1–93Q2	
	Univariate	Bivariate	Univariate	Bivariate
1. Boschen-Mills				
<95	−0.0756	−0.0049	−0.0074	−0.0074
	(0.0201)	(0.0156)	(0.0106)	(0.0175)
95–99	−0.1271	−0.0537	0.0397	0.0317
	(0.0445)	(0.0502)	(0.0269)	(0.0221)
>99	−0.0561	0.0082	0.1981	0.1620
	(0.0857)	(0.1092)	(0.0388)	(0.0361)
Small–Big	−0.0195	−0.0131	−0.2056	−0.1694
	(0.0897)	(0.1206)	(0.0407)	(0.0448)
2. Funds rate				
<95	−0.0193	−0.0004	−0.0260	−0.0256
	(0.0099)	(0.0070)	(0.0101)	(0.0133)
95–99	−0.0203	−0.0030	0.0367	0.0519
	(0.0232)	(0.0167)	(0.0216)	(0.0225)
>99	0.0473	0.0899	0.1936	0.2084
	(0.0258)	(0.0392)	(0.0318)	(0.0270)
Small–Big	−0.0665	−0.0903	−0.2196	−0.2339
	(0.0256)	(0.0456)	(0.0292)	(0.0294)
3. Bernanke-Mihov				
<95	−0.9554	1.6565	−1.8269	−2.9728
	(1.6555)	(0.4844)	(0.9415)	(0.8072)
95–99	−1.5482	1.2918	5.8490	8.2468
	(3.5492)	(2.2360)	(2.4472)	(2.9796)
>99	−0.1705	2.4275	15.2314	15.1635
	(3.4808)	(2.6893)	(2.3654)	(2.6942)
Small–Big	−0.7849	−0.7710	−17.0583	−18.1363
	(2.2147)	(2.7472)	(2.2301)	(2.4477)

Panel B: Total Loans	76Q1–85Q4		86Q1–93Q2	
	Univariate	Bivariate	Univariate	Bivariate
1. Boschen-Mills				
<95	−0.0417	−0.0095	0.0018	0.0008
	(0.0114)	(0.0160)	(0.0043)	(0.0055)
95–99	−0.0798	−0.0204	0.0334	0.0309
	(0.0189)	(0.0154)	(0.0106)	(0.0189)
>99	−0.0800	−0.0265	0.1449	0.1455
	(0.0469)	(0.0601)	(0.0249)	(0.0289)
Small–Big	0.0384	0.0170	−0.1430	−0.1447
	(0.0473)	(0.0609)	(0.0268)	(0.0312)
2. Funds rate				
<95	−0.0062	0.0015	−0.0079	−0.0135
	(0.0056)	(0.0048)	(0.0041)	(0.0062)
95–99	−0.0217	−0.0082	0.0175	0.0076
	(0.0089)	(0.0051)	(0.0158)	(0.0151)
>99	0.0050	0.0290	0.1004	0.1121
	(0.0162)	(0.0129)	(0.0180)	(0.0265)
Small–Big	−0.0113	−0.0275	−0.1083	−0.1256
	(0.0141)	(0.0167)	(0.0194)	(0.0282)
3. Bernanke-Mihov				
<95	−0.0246	1.6395	0.0169	−0.2564
	(0.9106)	(0.3712)	(0.3865)	(0.5633)
95–99	−1.4103	0.1350	1.9791	2.3618
	(1.7311)	(0.7047)	(1.2238)	(1.5434)
>99	1.1932	4.6248	10.0419	12.2748
	(3.1775)	(1.6549)	(1.1024)	(1.3265)
Small–Big	−1.2178	−2.9853	−10.025	−12.5311
	(2.3791)	(1.7011)	(1.0415)	(1.1759)

(b) Table 6: Split-sample results.

The evidence is stronger after 1986, when policy measurement is cleaner and Regulation Q concerns are less central.

9.6 Table 7: Economic significance

Read:

- The conservative small-bank C&I estimate accounts for 0.41 percent of small-bank C&I lending after a 100-basis-point funds-rate shock.
- The more aggressive small-bank minus big-bank differential accounts for 3.62 percent.
- These estimates compare with a 3.33 percent aggregate decline in small-bank C&I lending in the earlier Kashyap-Stein aggregate analysis.

Economic message: The lending-channel effect is economically meaningful, but the paper is careful not to claim a precise economywide magnitude.

TABLE 7—MOVEMENT IN AGGREGATE SMALL-BANK LENDING ACCOUNTED FOR BY CONSTRAINED BANKS FOUR QUARTERS AFTER A FEDERAL FUNDS-RATE SHOCK OF 100 BASIS POINTS

	Percentage change in lending due to constraints	Aggregate percentage change in lending
A. C&I loans		
1. Using univariate, small-bank sum of phi's	0.73	1.01[a]
2. Using bivariate, small-bank sum of phi's	0.41	3.33[b]
3. Using univariate, small–big bank differentials	2.90	1.01[a]
4. Using bivariate, small–big bank differentials	3.62	3.33[b]
B. Total loans		
1. Using univariate, small-bank sum of phi's	0.24	2.39[c]
2. Using bivariate, small-bank sum of phi's	0.13	3.15[d]
3. Using univariate, small–big bank differentials	0.95	2.39[c]
4. Using bivariate, small–big bank differentials	1.39	3.15[d]

Notes: The numbers in the first column are based on the two-step estimates reported in Table 3.

The numbers in the second column are drawn from Kashyap and Stein's (1995) estimates for the "small95" category as follows:

[a] Table 4, Panel 1;
[b] Table 4, Panel 2;
[c] Table 3, Panel 1;
[d] Table 3, Panel 2.

Table 7: Movement in aggregate small-bank lending accounted for by constrained banks.

The conservative estimates explain a nontrivial share of small-bank C&I lending movements after a funds-rate shock; aggressive estimates imply much larger effects.

10 Interpretation and identification logic

10.1 Why the result is hard to explain without a lending channel

A nonlending-channel explanation would need to argue that banks whose borrowers' loan demand is most sensitive to monetary policy systematically hold less liquidity. That is the heterogeneous-risk-aversion story. The paper argues that this explanation is not very plausible, and several facts go against it:

- The GDP coefficients are often positive, suggesting that liquid banks are more able to expand when demand rises.
- Very large banks show positive policy coefficients, which indicates that raw estimates are likely biased in the conservative direction for small banks.
- The results are strongest for C&I loans, where a bank lending channel should be easiest to detect.
- The results remain after several robustness checks that address loan composition, geography, time trends, and size aggregation.

10.2 What the paper proves and what it does not prove

What it establishes:

- Monetary policy has heterogeneous effects on bank lending depending on liquidity.
- This heterogeneity is strongest among small banks.
- These patterns are consistent with banks facing financing frictions that make internal liquidity valuable when policy tightens.

What remains uncertain:

- The exact aggregate size of the lending channel.
- How much borrowers can substitute from bank credit to nonbank credit when banks cut lending.
- Whether the real effects on investment, inventories, and employment are large.
- How the mechanism changes in more recent banking systems with different regulation, securitization, and funding structures.

11 Short summary

- The paper tests the bank lending channel using nearly one million bank-quarter observations.
- The key prediction is that lending should be more sensitive to monetary policy for banks with weaker liquidity buffers, especially among small banks.
- The baseline two-step method first estimates, quarter by quarter, how liquidity predicts loan growth within bank-size classes, then relates that liquidity sensitivity to monetary policy.
- The main C&I loan results show that liquidity matters more when policy is tight for small banks, but not for the largest banks.
- The same pattern appears for total loans but is weaker.
- Robustness checks support the result across alternative monetary indicators, one-step specifications, subsamples, and balance-sheet definitions.
- The effect is economically meaningful, though the paper is cautious about aggregate macro magnitudes.

12 Conclusion

12.1 Core conclusion

Kashyap and Stein provide strong evidence that monetary policy affects bank loan supply through the balance sheets of small banks. When policy tightens, lending becomes more dependent on liquid-asset buffers. Less-liquid small banks contract lending more than otherwise similar liquid small banks.

12.2 Economic intuition

The result is natural under financing frictions. Small banks cannot perfectly replace insured deposit funding with uninsured market funding. Securities holdings therefore act as a buffer stock. Banks with large buffers can absorb funding shocks without cutting loans; banks with small buffers cannot.

12.3 Incremental contribution revisited

The paper's main contribution is its cross-sectional identification of a lending channel. Instead of relying on aggregate correlations between monetary policy and bank lending, it asks whether the policy-lending relationship varies with bank liquidity in the direction predicted by theory. This makes the evidence much sharper than prior aggregate studies.

12.4 Limitations

The paper cannot fully quantify the aggregate macro effect because large-bank estimates appear biased and because the response of borrowers outside the banking system is not observed. A full macro assessment also requires knowing how easily bank-dependent firms substitute to nonbank credit.

12.5 Takeaway

The paper makes the existence of a U.S. bank lending channel hard to deny for the 1976-1993 period. Monetary policy is transmitted not only through open-market interest rates, but also through the balance-sheet capacity of banks, especially small and illiquid banks.